Walchand Institute of Technology, Solapur
(An Autonomous Institute)

Affiliated to
Punyashlok Ahilyadevi Holkar Solapur University, Solapur

Choice Based Credit System (CBCS)

Structure and Syllabus
for
Final Year B.Tech. Mechanical and Automation Engineering

W.E.F. 2026-27

Pr. S. B. Tuljapure
Chairman - Board of Studies

WIT SOLAPUR (Autonomous)

Dean (Academics)

Department of Mechanical Engineering

Vision

- To produce world class globally competent distinguished graduates/ post graduates/ doctoral, Mechanical Engineers on the basis of their capabilities, dedication and work ethic and continuously strive towards societal development.

Mission

- **M1:** To impart quality Mechanical Engineering education in accordance with the needs of the society.
- **M2:** To produce globally competent Mechanical Engineers through research, industry institute interaction.
- **M3:** To help Mechanical Engineering graduates to implement their acquired engineering knowledge for society and community development.

Mechanical and Automation Engineering

Program Educational Objectives (PEOs)

- Graduate will excel in professional career in the field of Mechanical and Automation Engineering.
- Graduate will exhibit strong fundamentals required to pursue higher education and continue professional development in emerging technology in Mechanical and Automation Engineering.
- Graduate will adhere to ethics develop team spirit and effective communication skills to be successful leaders with a holistic approach to societal and environmental issues with professional conduct.

Knowledge and Attitude Profile (WK)

WK1	A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences.
WK2	Conceptually-based mathematics, numerical analysis, data analysis, statistics and formal aspects of computer and information science to support detailed analysis and modelling applicable to the discipline.
WK3	A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.
WK4	Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline.
WK5	Knowledge, including efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, and similar concepts, that supports engineering design and operations in a practice area.
WK6	Knowledge of engineering practice (technology) in the practice areas in the engineering discipline.
WK7	Knowledge of the role of engineering in society and identified issues in engineering practice in the discipline, such as the professional responsibility of an engineer to public safety and sustainable development.
WK8	Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues.
WK9	Ethics, inclusive behaviour and conduct. Knowledge of professional ethics, responsibilities, and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability etc. with mutual understanding and respect, and of inclusive attitudes.

Program Outcomes (POs)	
PO 1	**Engineering Knowledge:** Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.
PO 2	**Problem Analysis:** Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)
PO 3	**Design/Development of Solutions:** Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)
PO 4	**Conduct Investigations of Complex Problems:** Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).
PO 5	**Engineering Tool Usage:** Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)
PO 6	**The Engineer and The World:** Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).
PO 7	**Ethics:** Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)
PO 8	**Individual and Collaborative Team work:** Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
PO 9	**Communication:** Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning difference
PO 10	**Project Management and Finance:** Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.
PO 11	**Life-long Learning:** Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and

	emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

Program Specific Outcomes (PSOs)

1. Design high quality mechanical and automation engineering equipments for the modern industry and society.
2. Implement manufacturing processes for mechanical and automation engineering equipment.

Department of Mechanical Engineering

Legends Used

L	Lecture Hours / week
T	Tutorial Hours / week
P	Practical Hours / week
FA	Formative Assessment
SA	Summative Assessment
ESE	End Semester Examination
ISE	In Semester Evaluation
ICA	Internal Continuous Assessment
POE	Practical and Oral Exam
OE	Oral Exam
MOOC	Massive Open Online Course
HSS	Humanity and Social Science
NPTEL	National Programme on Technology Enhanced Learning
F.Y.	First Year
S.Y.	Second Year
T.Y.	Third Year
B. Tech.	Bachelor of Technology

Mechanical and Automation Engineering

Course Code Format

2	3	M	A	U/P	2	C	C	1	T/L
Year of Syllabus revision		Program Code		U-Under Graduate P-Post Graduate	Semester No./ Year 1/2/3/…8	Course Type		Course Serial No 1-9	Theory, L-Lab session P- Programming

Program Code	
MA	Mechanical and Automation Engineering
Course Type	
BS	Basic Science
ES	Engineering Science
HU	Humanities & Social Science
MC	Mandatory Course
CC	Core Compulsory Course
SN*	Self-Learning **N*** *indicates the serial number of electives offered in the respective category*
EN*	Core Elective **N*** *indicates the serial number of electives offered in the respective category*
ON*	Open Elective **N*** *indicates the serial number of electives offered in the respective category*
SK	Skill Based Course
SM	Seminar
MP	Mini project
PR	Project
IN	Internship
RM	Research Methodology

Sample Course Code	
23MAU7CC1T	Refrigeration and Air Conditioning

Walchand Institute of Technology, Solapur

(An Autonomous Institute Affiliated to P.A.H. Solapur University, Solapur)

Structure of Final Year B. Tech. Mechanical and Automation Engineering

Mechanical and Automation Engineering

B. Tech. Semester VII

Course Code	Name of Course	Engagement Hours			Credits	SA		FA		Total
		L	T	P		Theory	OE/ POE	ISE	ICA	
23MAU7CC1T	Refrigeration & Air Conditioning	3			3	60		40		100
23MAU7EN*2T	Programme Elective Course IV	3			3	60		40		100
23MAU7EN*3T	Programme Elective Course V	2			2	60		40		100
23MAU7RM4T	Research Methodology & Intellectual Property Rights	2			2			50		50
23##U7MD5T	Multidisciplinary Minor V	3			3	60		40		100
23##U7MD5A	Multidisciplinary Minor V (Tutorial)		1		1				25	25
	Subtotal	**13**	**1**		**14**	**240**		**210**	**25**	**475**
Laboratory Courses										
23MAU7CC1L	Refrigeration & Air Conditioning			2	1		25		25	50
23MAU7EN*2L	Programme Elective Course IV			2	1		50		25	75
23MAU7EN*3L	Programme Elective Course V			2	1				25	25
23MAU7RM4L	Research Methodology & IPR			2	1				25	25
23MAU7PR4L	Capstone Project-I			8	4		50		100	150
	Subtotal			**16**	**8**		**125**		**200**	**325**
	Grand Total	**13**	**1**	**16**	**22**	**240**	**125**	**210**	**225**	**800**

Note: - N* indicates the serial number of electives offered in the respective category
indicates program code of offering Programme

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Programme Elective Course –IV (Semester –VII)

Course Code	Course Name
23MAU7E12T	Design of Mechatronic Systems
23MAU7E22T	Supply Chain Management
23MAU7E32T	Computer Integrated Manufacturing

Programme Elective Course –V (Semester –VII)

Course Code	Course Name
23MAU7E13T	Piping Design
23MAU7E23T	Industrial Engineering
23MAU7E33T	Industrial Robotics

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Mechanical and Automation Engineering

B. Tech. Semester VIII

Course Code	Name of Course	Engagement Hours			Credits	SA		FA		Total
		L	T	P		Theory	OE/ POE	ISE	ICA	
23MAU8EN*1T	Power Plant Engineering / Programme Course (MOOC/Self Learning) #	2			2	50				50
23MAU8EN*2T	Programme Elective Course VI / (MOOC/Self Learning) ##	2			2	50				50
	Subtotal	4			4	100				100
Laboratory Courses										
23MAU8IN3L	Internship			20	10		50		100	150
23MAU8PR4L	Capstone Project-II			04	02		50		100	150
	Subtotal			24	12		**100**		200	300
	Grand Total	**4**		**24**	**16**	**100**	**100**		**200**	**400**

Note: - N* indicates the serial number of electives offered in the respective category.

The NPTEL course (MOOC) of eight weeks duration for programme course is already announced by the BoS Chairman at the start of the semester VI & this course is Power Plant Engineering (MOOC) with course code 23MAU8E11T.

Students shall register, complete the course & appear for examination conducted by NPTEL before the end of semester VIII. The passing certificate should be submitted to the department for the credit transfer.

In case student has not passed the examination conducted by the NPTEL before the end of semester VIII, he/she can appear for the examination conducted by the institute at the end of semester VIII. In that case the course will be treated as Power Plant Engineering (Self learning) instead of Power Plant Engineering (MOOC) & course code will be 23MAU8E21T

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The list of NPTEL courses (MOOCs) of eight weeks duration for programme elective course is already announced by the BoS Chairman at the start of the semester VI & these courses with course codes are as given below.

Programme Elective Course –VI (Semester –VIII) MOOC

Course Code	Course Name
23MAU8E12T	Welding Application Technology
23MAU8E22T	Advances in Welding and Joining Technologies
23MAU8E32T	Work System Design

Students shall register, complete the course & appear for examination conducted by NPTEL before the end of semester VIII. The passing certificate should be submitted to the department for credit transfer.

In case student has not passed the examination conducted by NPTEL before the end of semester VIII, he/she can appear for the examination conducted by the institute at the end of semester VIII. In that case the course will be treated as Self learning instead of MOOC & course code will be as below.

Programme Elective Course –VI (Semester –VIII) Self Learning

Course Code	Course Name
23MAU8E42T	Welding Application Technology
23MAU8E52T	Advances in Welding and Joining Technologies
23MAU8E62T	Work System Design

- **Internship should be of minimum 90 days.**
- **Internship may be offered by**
    - Industry at their premises.
    - Industry at the institute campus.
    - Institute jointly with the research funded agency/ industry.

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- **Multidisciplinary Minor (MDM) Courses**

Sr. No.	MDM Program	MDM I (Sem III)	MDM II (Sem IV)	MDM III (Sem V)	MDM IV (Sem VI)	MDM V (Sem VII)
1	Computer Science and Engineering	Operating System	UI Technologies	Software Engineering	Big Data Technologies	Software Testing and Quality Assurance
2	Information Technology	Principles of Operating Systems	Web UI and UX Technology	Software Engineering Principles	Data Preprocessing and Visualization	Cyber Security
3	Civil Engineering	Smart Buildings	Geoinformatics	Environmental Impact Assessment	Infrastructural Systems	Disaster Preparedness and planning
4	Electronics and Telecommunication Engineering	Fundamentals of Electronic Circuits	Electronics Design and Prototyping	Introduction to Embedded Systems	Fundamentals of Communication Techniques	Enclosure and Communication Design for IoT
5	Electronics and Computer Engineering	Analog Electronics	Digital Electronics	Microcontrollers and Peripherals	Advanced Controllers and Interfacing	Electronic System Design

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	WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR **(An Autonomous Institute)** **Final Year B.Tech. (Mechanical and Automation Engineering), Semester-VII**

23MAU7CC1T: Refrigeration and Air-Conditioning

Teaching Scheme		**Examination Scheme**	
Lectures	3 Hours/week	**ESE**	60 Marks
Practical	2 Hours/week	**ISE**	40 Marks
Credits	4	**ICA**	25 Marks
		OE	25 Marks

Introduction:

The course consists of different refrigeration cycles such as Air refrigeration cycle, Vapour Compression cycle, Vapour absorption cycle. It also covers properties of refrigerants and various alternative refrigerants and understanding of psychrometric chart and psychrometric processes used for the purpose of air- conditioning. Further, the comfort air-conditioning and cooling load calculations are also addressed in this course.

Course Prerequisite:

Applied Thermodynamics, Heat Transfer

Course Objectives:

1. To learn the fundamental principles and different methods of refrigeration and air conditioning.
2. Understand basic refrigeration processes
3. Study of different refrigerants with respect to properties, applications and environmental issues.
4. Understand the basic air conditioning processes on psychometric charts, calculate cooling load for its applications in comfort and industrial air conditioning.
5. To equip learners with the skills needed to design and analyze refrigeration and air conditioning components and systems.

Course Outcomes:

After completing the course, students will be able to

1. Evaluate performance of various types of refrigeration systems
2. Select appropriate refrigerant considering necessary properties
3. Use Psychrometric chart and tables and analyse psychrometric process for obtaining required air conditions.
4. Analyse the factors influencing cooling load and perform its calculations.

Unit – I	**Basic Refrigeration Cycles and Refrigerants**	**6 Hours**

A) Introduction
Refrigeration, Units of refrigeration, Applications of refrigeration, Reversed Carnot cycle with vapour as refrigerant, refrigerator and heat pump, Calculation of COP.
B) Refrigerant

Classification, Desirable Properties, Nomenclature of Refrigerants, Selection of refrigerant, Secondary refrigerants, Ozone depletion potential (ODP) and Global warming potential (GWP), Total equivalent warming impact (TEWI), Alternative Refrigerant		
Unit – II	**Vapour Compression Systems**	**8 Hours**
Working of simple vapour compression system, representation of different vapour compression cycle (VCC) on T-s and P-h diagram, Sub cooling, Superheating, liquid-suction heat exchanger, Analysis and Performance calculations of above cycles. Effect of operating parameters on performance of VCC, actual VCC, methods of improving COP, Multipressure systems: Flash gas removal, Flash inter cooling, Compound compression with intercooling, Multiple evaporator systems, Cascade refrigeration system, Introduction to cryogenics Limitations of vapour compression systems for the production of low temperature, Cascade Refrigeration System, Linde System for liquefaction of air. Applications of Cryogenics.		
Unit – III	**Vapour Absorption Systems**	**5 Hours**
Working of simple vapour absorption system (VAS), Practical vapour absorption system, desirable properties of binary mixture (aqua-ammonia), COP of an ideal Vapour Absorption Refrigeration System, Li-Br absorption system, three fluid system (Electrolux refrigeration), applications of VARS, comparison between VCRS and VARS.		
Unit – IV	**Psychrometry**	**11 Hours**
Dalton's law of partial pressure, Psychometrics properties of moist air, Enthalpy of moist air, use of psychometric tables and Charts, Psychometrics Processes, Combinations and Calculations, SHF, BPF, ADP, Coil condition line, Air washer, Human Comfort condition, Thermodynamics of human body, comfort and comfort chart, factors affecting human comfort, concept of infiltration and ventilation, indoor air quality requirements.		
Unit – V	**Heating and Cooling Load Calculations**	**6 Hours**
Enumeration and brief explanation of the factors forming load on refrigeration and air conditioning systems, Ventilation requirements according to ASHRAE std. 62.1, Inside and Outside Design conditions, Cooling load calculations, Load analysis by RSHF, GSHF, and ESHF		
Unit – VI	**Air Conditioning and Air Distribution Systems**	**4 Hours**
A) Room air conditioning, Chilled water systems, DX systems, Comparison between Chilled water and DX systems, Air handling unit, Fan coil unit, Desert coolers, Air-washer, Industrial applications B) Classification of ducts, pressure in ducts, flow through duct, equivalent diameter, Methods of duct system design: equal friction, velocity reduction, static regain method, types of fans used air conditioning applications, grills, registers, diffusers.		
ICA consists of minimum 8 practical/assignments from the below list 1. Study of Refrigeration methods 2. Study of charging, leak testing of refrigeration systems 3. Trial on refrigeration test rig. 4. Trial on heat pump test rig. 5. Trial on Vapour Absorption system 6. Trial on Air conditioning tutor 7. Calculation of cooling load for given space drawing 8. Visit to Refrigeration plant or Central Air Conditioning plant 9. Study of ASHRAE codes. 10. Study of green buliding/net zero building.		

Text Books
1. C. P. Arora, Refrigeration & Air Conditioning, McGraw Hill 2. Manohar Prasad, Refrigeration & Air Conditioning, New Age International Publications
Reference Books
1. Stocker, Refrigeration & Air Conditioning, McGraw Hill 2. Roy J Dossat, Principles of Refrigeration, Pearson Publications 3. W. P. Jones, Air Conditioning Applications & design, B. H. Publications
e-Resources
https://onlinecourses.nptel.ac.in/e-learning/preview/noc21_me85

WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR **(An Autonomous Institute)** **Final Year B.Tech. (Mechanical and Automation Engineering), Semester-VII**			

Programme Elective Course IV
23MAU7E12T: Design of Mechatronic Systems

Teaching Scheme		**Examination Scheme**	
Lectures	3 Hours/week	**ESE**	60 Marks
Practical	2 Hours/week	**ISE**	40 Marks
Credits	4	**ICA**	25 Marks
		OE	50 Marks

Introduction:

Mechatronic system design strives to integrate mechanical, electronic, optical, and computer technologies in order to create "optimal" products and processes. Basic concepts and fundamental principles will be reviewed in this course. Students will develop the knowledge and skills necessary to adopt an interdisciplinary approach to mechatronic system design through the lectures, hands-on laboratory assignments, and a term design project.

Course Prerequisite:

Prior completion of a course in Mechatronics, along with foundational knowledge of Basic Electrical and Electronics Engineering, is required.

Course Objectives:

1. To describe core features and peripheral features of Microprocessor and Microcontroller.
2. To provide an introduction to microcontroller families and details of RISC and CISC microcontrollers.
3. To equip students with advanced knowledge in programmable logic controller technology to solve problems related to mechatronic systems.
4. Design mechatronic systems for real-world industrial and consumer applications such as pick-and-place robots, car parking barriers, washing machines, and camera systems.

Course Outcomes:

After completing this course, student shall be able to –

1. Program microcontrollers in assembly/and or C/C++/Python/Java to demonstrate interfacing with sensors and actuators.
2. Program PLCs using ladder logic (both on simulators and actual hardware).
3. Build and program a mechatronic system which will accept data from input and sensors and control an output/actuator using any microprocessor/ microcontroller board (Arduino or Raspberry Pi can also be used)
4. Interpret, analyse, and model real-world mechatronic systems involving motion control, thermal control, and sensor-actuator integration using practical case studies.

Unit – I	**Microprocessor Fundamentals**	**6 Hours**
Fundamentals of a computing system, microprocessor definition, memory and its types, Von Neumann vs Harvard Architecture, RISC/CISC, Introduction to 8085 microprocessor, 8085 architecture and pin layout, memory, 8085 programming and instruction set, addressing and interfacing, interrupts, timing diagram		
Unit – II	**Microcontroller Fundamentals**	**6 Hours**
Microcontrollers, 8051 and Arduino and Raspberry PI development boards architecture and pin layout, interfacing and programming, instruction set, addressing modes, special registers, Timers/Counters, Serial Communication, Interrupts, Simple assembly programs.		
Unit – III	**Programmable Logic Controller**	**8 Hours**
PLC architecture, I/O Processing, NPN/PNP sourcing and sinking, Ladder Diagrams, SFC, FBD , Internal Relays, Jump and Call, Timers, Counters, Shift Registers and Data Handling, Programs, for temperature control, sequencing etc., PLC Vs. PC based systems, top manufacturers		
Unit – IV	**Introduction to embedded systems**	**6 Hours**
Microcontroller development boards and embedded programming platforms, Instruction set architecture of ARM microcontroller, and assembly language programming, Introduction to Internet of Things, smart home concepts		
Unit – V	**Analysis of Control systems in State Space**	**6 Hours**
State space representation of transfer function systems, transformation of system models with MATLAB, Solution of time invariant state equation, controllability		
Unit – VI	**Case Studies on Design of Mechatronic systems**	**8 Hours**
Motion control using DC Motor, AC Motor and Servomotor - Temperature control of hot/cold reservoir - Pick and place robot - Car parking barriers – Motion and temperature control of washing machine - Auto focus camera, exposure control		
ICA consists of minimum 8 practical based on curriculum. Recommended practical: 1. An assignment on sensors and filter circuits 2. Demonstration of I2C communication using Arduino UNO 3. Interfacing of IR sensor module with Raspberry Pi 400 4. Interfacing of sound sensor module with Raspberry Pi 400 5. 8085 Microprocessor programming using simulator 6. 8051 Microcontroller programming using simulator 7. PLC programming using ladder logic on a simulator and hardware		

8. Speed control of DC motor using Raspberry Pi and Arduino

9. Interfacing of ultrasonic sensor module with Raspberry Pi 400

10. Interfacing of servo motor with Raspberry Pi 400

Text Books

1. W. Bolton, Mechatronics: Electronic Control Systems in Mechanical and Electrical Engineering, Pearson Publishing, 7th Edition
2. W. Bolton, Programmable logic Controllers, Pearson Publishing, 4th Edition
3. Mazidi, 8051 Microcontroller, Prentice Hall, 3rd Edition

Reference Books

1. Bishop et.al, Handbook of Mechatronics, CRC Press, 2nd Edition
2. Gaonkar Ramesh, The 8085 microprocessor, Penram International Publishing, 2nd Edition
3. Bradley, D.Dawson, N.C. Burd and A.J. Loader, Mechatronics: Electronics in Products and Processes, CRC Press 1991, First Indian print 2010.
4. De Silva, Mechatronics: A Foundation Course, Taylor & Francis, Indian Reprint, 2013.
5. F. Vahid and T. Givargis, Embedded System Design: A Unified Hardware/Software Introduction, Wiley India Pvt. Ltd., 2002.
6. A.N. Sloss, D. Symes and C. Wright, ARM System Developer's Guide: Design and Optimizing System Software, Morgan Kaufman Publishers, 2004.
7. W. Wolf, Computers as Components: Principles of Embedded Computing System Design, Morgan Kaufman Publishers, 2008.

e-Resources

https://www.youtube.com/watch?v=LX2oxCz8cU4&list=PLOzRYVm0a65dAI5mqWdoXKExqzWn0pK-j

WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR

(An Autonomous Institute)

Final Year B.Tech. (Mechanical and Automation Engineering), Semester-VII

Programme Elective Course IV

23MAU7E22T: Supply Chain Management

Teaching Scheme		**Examination Scheme**	
Lectures	3 Hours/week	**ESE**	60 Marks
Practical	2 Hours/week	**ISE**	40 Marks
Credits	4	**ICA**	25 Marks
		OE	50 Marks

Introduction:

Supply Chain Management (SCM) is a critical discipline encompassing the strategic coordination and integration of activities across organizations to improve and optimize the flow of goods, services, information, and finances from raw material suppliers through production to the end customer. This course provides a comprehensive overview of SCM principles, strategies, and practices essential for effective management of modern supply chains.

Course Prerequisite:

No special prerequisite is needed for this course

Course Objectives:

1. To understand the fundamentals and advance concept of supply chain management
2. To describe the key components of supply chains: manufacturing and operations management, distribution and logistics, and inventory management.
3. To study the importance of coordination and integration for supply chain performance
4. To understand the role of metrics in evaluating supply chain effectiveness and efficiency.
5. To study sustainability and risk management within supply chains.

Course Outcomes:

After completing the course, students will be able to

1. Analyze the components and drivers of a supply chain and their impact on organizational performance.
2. Apply quantitative models such as EOQ and inventory control systems to solve real-world materials management and logistics problems.
3. Evaluate supply chain performance using key performance indicators (KPIs) and propose improvements in supply chain coordination and integration.
4. Demonstrate the use of IT tools such as ERP, SCM software, or data analytics to enhance supply chain visibility and efficiency.

Unit – I	**Introduction to SCM**	**4 Hours**

Definitions of supply chain management and logistics, evolution of supply chain management, objectives of SCM, importance of SCM, activities of supply chain management, process view in supply chain, supply chain network and flow of materials and information, supply chain drivers, decision phases in supply chain

Unit – II	**Supply Chain components and strategy**	**4 Hours**

Supply chain components, manufacturing and operations management, distribution and logistics, inventory management, supply chain strategy-differentiation vs cost leadership global supply chain considerations, risk management and resilience		
Unit – III	**Materials Management**	**4 Hours**
Definition and scope of materials management, importance of materials management in supply chains, classification of materials, procurement, purchasing policies, vendor development and evaluation. inventory control systems of stock replenishment, cost elements, EOQ and its derivative modules		
Unit – IV	**Logistics and Transportation**	**3 Hours**
Modes of transportation (road, rail, sea, air), routing and scheduling, third-party logistics (3PL) and fourth-party logistics (4PL), procurement and supplier relationship management, supplier selection and evaluation, supplier collaboration and partnerships		
Unit – V	**Supply Chain Coordination and Integration**	**4 Hours**
Definition and scope of supply chain coordination and integration, importance of coordination and integration for supply chain performance, information sharing in supply chains, Collaborative Planning, Forecasting, and Replenishment (CPFR) process integration in supply chains, supply chain collaboration platforms		
Unit – VI	**Supply Chain Performance Metrics**	**4 Hours**
Definition and significance of supply chain performance metrics, role of metrics in evaluating supply chain effectiveness and efficiency, types of performance metrics, operational, financial and strategy, Key Performance Indicators (KPIS) in supply chains, operational performance metrics, financial performance metrics, customer service metrics supply chain risk management metrics, sustainability metrics in supply chain		
Unit – VII	**IT in Supply Chain**	**2 Hours**
Importance of it in supply chain efficiency and effectiveness, evolution of it in supply chain management, supply chain information system- Enterprise Resource Planning (ERP) systems Supply Chain Management (SCM) software Warehouse Management Systems (WMS) Transportation Management Systems (TMS), data analytics and supply chain visibility		
Unit – VIII	**Emerging Trends in Supply Chain**	**4 Hours**
Introduction to emerging trends in supply chain management, digital transformation in supply chains Internet of Things (IOT) in supply chains, blockchain technology in supply chains, artificial intelligence (ai) and Machine Learning (ML) in supply chains, sustainable supply chain management circular economy in supply chains, supply chain collaboration and partnership		

ICA consists of minimum 8 practical/assignments from the below list

1. A real-world case study regarding the evolution and strategic importance of supply chain management
2. Draw and explain the supply chain network and process flow for a selected manufactured product
3. Calculate EOQ, reorder point, and safety stock for a given inventory management scenario
4. Classify materials using ABC analysis and develop a vendor evaluation matrix for supplier selection
5. Solve a vehicle routing or scheduling problem using optimization techniques or routing software
6. Apply a multi-criteria decision-making method to select the most suitable supplier from a given dataset
7. Analyze Key Performance Indicators (KPIS) of a supply chain and suggest improvements

8. Simulate the Collaborative Planning, Forecasting, and Replenishment (CPFR) process for a product line
9. Demonstrate the use of ERP or SCM software to track and manage supply chain operations
10. Prepare a report on the role of it and data analytics in enhancing supply chain visibility and decision-making

Text Books

1. Sunil Chopra and Peter Meindl, Supply Chain Management: Strategy, Planning, and Operation, Pearson Publications
2. Garg B.L., Karadia, R., Agarwal, F. and Agarwal, U.K., 2002, An introduction to Research Methodology, RBSA Publishers
3. Mr. Ishanka Saikia, Mr. V. Anandaraj, Dr. S. Ramachandran, S. Kumaran, Supply Chain & Logistics Management, Airwalk Publications

Reference Books

1. Sunil Sharma, Supply Chain Management- Concepts, Practices, and Implementation, Oxford University Press.
2. F. Robert Jacobs and Richard B. Chase, Operations and Supply Chain Management, McGraw Hill Education
3. Cecil C. Bozarth and Robert B. Handfield, Introduction to Operations and Supply Chain Management, Pearson Publications

e-Resources

https://archive.nptel.ac.in/courses/110/ Pearson106/110106045/

WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR
(An Autonomous Institute)
Final Year B.Tech. (Mechanical and Automation Engineering), Semester-VII

Programme Elective Course IV
23MAU7E32T: Computer Integrated Manufacturing

Teaching Scheme		**Examination Scheme**	
Lectures	3 Hours/week	**ESE**	60 Marks
Practical	2 Hours/week	**ISE**	40 Marks
Credits	4	**ICA**	25 Marks
		OE	50 Marks

Introduction:

Computer Integrated Manufacturing (CIM) is a comprehensive approach to manufacturing where computers control the entire production process—from product design and planning to manufacturing, quality control, and distribution. This course provides a strong foundation in the principles, technologies, and applications that integrate design, manufacturing, automation, and digital systems into a unified production environment.

Course Prerequisite:

No pre-requisite required

Course Objectives:

1. To develop a comprehensive understanding of Computer Integrated Manufacturing (CIM) systems, including their basic principles, components, and functions.
2. To acquire proficiency in utilizing computer-aided design (CAD), computer-aided manufacturing (CAM), and computer-aided engineering (CAE) tools for designing, analyzing, and optimizing manufacturing processes.
3. To explore the role of automation and robotics in the manufacturing industry, including programming and controlling industrial robots and understanding their integration within CIM systems.
4. To evaluate the benefits and challenges associated with implementing CIM systems, analyze their impact on manufacturing processes, and assess the potential for optimization and improvement.

Course Outcomes:

After completing the course, students will be able to

1. Describe the principles, components, and functions of Computer Integrated Manufacturing (CIM) systems and interpret their role in modern manufacturing processes.
2. Apply CAD, CAM, and CAE tools to design, analyze, and optimize manufacturing processes.
3. Demonstrate the programming and control of industrial robots and integrate robotic systems within CIM-based manufacturing environments.
4. Analyze the benefits and challenges of CIM implementation and evaluate strategies for improving manufacturing productivity, quality, and automation.

Unit – I	**Introduction to Computer Integrated Manufacturing**	**7 Hours**
Need of Computer Integrated Manufacturing (C.I.M), Components of C.I.M, Product cycle– General Process, Industry 4.0-Digital Manufacturing.		
Unit – II	**CAD and Computer Graphics Software**	**7 Hours**
Fundamentals of CAD, Product design and CAD, Geometric modelling, 2D Geometrical Transformations and types, Software configuration, functions of graphics package, Features of 3D modelling.		
Unit – III	**Cellular Manufacturing**	**6 Hours**
Computer Aided Manufacturing, Group Technology, Part families, Parts classification and coding, Computer Aided Process Planning and its types, Machine cells.		
Unit – IV	**Flexible Manufacturing System (FMS) And Automated Guided Vehicle System (AGV)**	**7 Hours**
Types of Flexibility - FMS – FMS Components – FMS Application & Benefits – FMS Guided Vehicle System (AGVS) – AGVS Application – Vehicle Guidance technology – Vehicle Management & Safety.		
Unit – V	**Automation and Robotics in Manufacturing**	**7 Hours**
Introduction to automation systems, Types of industrial robots and their applications, Robot programming and control, Integration of robots into CIM systems.		
Unit – VI	**Additive Manufacturing (AM)**	**6 Hours**
Basic principles of additive manufacturing, slicing CAD models for AM, classification of AM processes, base materials used in AM, applications of AM.		

ICA consists of minimum 8 practical/assignments from the below list

1. Introduction to Computer Integrated Manufacturing and Industry 4.0
2. CAD Fundamentals and 2D Geometrical Transformations
3. 3D Geometric Modeling and Product Design using CAD
4. Computer Aided Manufacturing and CNC Programming
5. Group Technology and Part Classification & Coding
6. Case study on Computer Aided Process Planning (CAPP)
7. Demonstration on Flexible Manufacturing Systems (FMS)
8. Design an AGV layout for material handling in a manufacturing cell
9. Automation and Robotics Integration in CIM
10. Demonstration of Additive Manufacturing and Digital Production Processes

Text Books

1. James A. Rehg and Henry W. Kraebber, Computer Integrated Manufacturing, Pearson College Div
2. P. N. Rao, CAD/CAM: Principles and Applications, McGraw Hill Education

Reference Books

1. Nikolaus Correll and Bradley Hayes, Introduction to Autonomous Robots: Kinematics, Perception, Localization and Planning, The MIT Press
2. Kamrani A.K. and Nasr E.A., Rapid Prototyping: Theory and practice, Springer,
3. Hilton P.D. and Jacobs P.F.,Rapid Tooling: Technologies and Industrial

4. Applications, CRC press 5. Mikell P. Groover, Automation, Production Systems, and Computer-Integrated Manufacturing, PHI
e-Resources
https://nptel.ac.in/courses/112104289

Programme Elective Course V
23MAU7E13T: Piping Design

Teaching Scheme		Examination Scheme	
Lectures	2 Hours/week	**ESE**	60 Marks
Practical	2 Hours/week	**ISE**	40 Marks
Credits	3	**ICA**	25 Marks

Introduction:

Piping design and engineering is a key area in various streams of engineering. Piping and accessories constitute over 25% of the total capital investment in the chemical process industry, petroleum and petrochemical industry, pharmaceutical industry, power plants, and so on. The present course is intended to familiarize undergraduate students about the fundamental design aspects of piping components and their applications in process industries.

Course Prerequisite:

The student should have basic understanding of fluid mechanics, engineering and mechanical properties associated with the material.

Course Objectives:

1. To introduce the fundamentals of piping systems including pipe classification, materials, codes and standards, fabrication practices, and vibration control.
2. To understand the design calculations such as pipe sizing, pressure drop estimation, pump selection, NPSH evaluation, and flow measurement.
3. To provide knowledge of piping components including fittings, valves, flanges, gaskets, steam traps, and analysis of single-phase and two-phase flow including water hammer effects.
4. To enable students to perform mechanical design of piping systems considering operating conditions, stress analysis, and pipe thickness calculations for internal and external pressure.
5. To familiarize students with pipe supports, thermal expansion analysis, expansion joints, and interpretation of PFD and P&ID for process equipment.

Course Outcomes:

After completing the course, students will be able to

1. Explain fundamentals of piping systems including pipe classification, codes & standards, material selection, mechanical properties, and vibration control.
2. Perform design calculations for piping systems including pipe sizing, pressure drop estimation, pump selection, NPSH analysis, and flow measurement.
3. Select and evaluate piping components such as fittings, valves, flanges, gaskets, steam traps, and analyze single-phase and two-phase flow including water hammer effects.
4. Design piping systems considering operating and design conditions, calculate stresses, and determine pipe thickness for internal and external pressure applications.
5. Calculate the dimensions of pipe support systems, analyze thermal expansion, select expansion joints, and interpret PFD and P&ID diagrams for major process equipment.

Unit – I	**Fundamentals of piping**	**4 Hours**
Classification of pipe, Codes and standards, Pipe Fabrication, vibration, its prevention and control in piping systems, Mechanical Properties of material, schedule number, Piping materials and selection		
Unit – II	**Design calculations for piping**	**6 Hours**
Determination of pipe size, Calculation of pressure drop in pipe, Equivalent length of pipe line for fittings and valves, Energy losses in pipe line, Different types of pumps and their selection criteria, NPSHA & NPSHR, Power required by pump, Calculation of fluid flow in pipe line.		
Unit – III	**Piping component and Flow through pipe line**	**5 Hours**
Types of Fitting, Different types of flanges and gaskets, their selection criteria and applications, Different types of valves, their selection criteria and applications, Determination of valve size, Steam separators and steam traps, calculation of pressure drop for two phase flow through pipe line by using Lockhart and Martinelli correlations, Piping drainage and water hammer in process plant, Calculations for water hammer in pipeline.		
Unit – IV	**Mechanical design of piping**	**7 Hours**
Operating pressure and temperature, Design Pressure & Design Temperature for Piping Systems, Design equation for longitudinal, hoop and allowable stresses, Determination of thickness required by steel pipe for withstanding internal and external pressure, Determination of thickness required by jacketed steel pipe for withstanding external pressure.		
Unit – V	**Pipe supports and P & I diagram**	**6 Hours**
Functions of supports and selection, Types of loads, different types of piping supports, Determination of support location, Maximum span between the supports suggested by ASME B 31.1, Thermal expansion in pipe line, Different types of expansion joints and their applications, Difference between a PFD and P&ID, Typical P&I diagrams for pumps, distillation column, Reactors and Shell and tube heat exchanger.		
ICA consists of minimum 8 practical/assignments from the below list 1. Prediction/Estimation of various physical properties such as density, viscosity, surface tension, specific heat, thermal conductivity of fluid. 2. Finding suitable material of construction for handling various chemicals. 3. Determination of pipe size and pressure drop for various fittings in the pipe line. 4. Determination of Power required by centrifugal pump. 5. Design of pipe flow measurement devices. 6. Determination of valve size for pipe line. 7. Determination of pressure drop for two phase flow through pipe line. 8. Determination of water hammer in pipe line. 9. Design of pipe subjected to internal and external pressure. 10. Design of jacketed steel pipe under external pressure. 11. Design of piping support. 12. P&I diagrams for pumps, distillation column, Reactors, Shell and tube heat exchanger.		

Text Books

1. A. K. Roy, Fundamentals of Piping Design (A Design and Data Book), Khanna Publishing House, 2022
2. Roy A. Parisher and Robert A. Rhea, Pipe drafting and design, Gulf Professional Publishing, 2022
3. Nayyar M.L., Piping Handbook, Tata McGraw Hill Publication, 2000.

Reference Books

1. Perry R.H., Chemical Engineers' Handbook, McGraw-Hill, 2009.
2. Coulson J.M, Richardson J.F and Sinnott, R.K., Coulson and Richardson's Chemical Engineering, Vol. 6, 4th Edition, Elsevier, New Delhi, 2006.
3. McCabe W.L, Smith J.C, Harriott P., Unit Operations of Chemical Engineering, Mc Graw Hill Publication, 2021

e-Resources

1) http://www.piping-engineering.com
2) https://www.eng-tips.com
3) http://www.pipingstudy.com
4) https://www.whatispiping.com
5) http://www.wermac.org
6) https://www.engineeringtoolbox.com
7) https://www.pipingengineer.org

WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR
(An Autonomous Institute)
Final Year B.Tech. (Mechanical and Automation Engineering), Semester-VII

Programme Elective Course V
23MAU7E23T: Industrial Engineering

Teaching Scheme		**Examination Scheme**	
Lectures	2 Hours/week	**ESE**	60 Marks
Practical	2 Hours/week	**ISE**	40 Marks
Credits	3	**ICA**	25 Marks

Introduction:

Industrial Engineering focuses on the design, improvement, and installation of integrated systems involving people, materials, information, equipment, and energy. It combines knowledge from mathematical, physical, and social sciences with engineering principles and design methods to analyze, predict, and evaluate the performance of such systems.

This course introduces the fundamental concepts of Industrial Engineering and highlights their applications in enhancing productivity in both manufacturing and service sectors. Students will learn important techniques such as method study and work measurement, along with job evaluation and merit rating, to improve organizational efficiency.

The course also provides knowledge of plant layout, facility location, safety practices, and ergonomic considerations, all of which contribute to creating effective work environments and achieving higher productivity.

Course Prerequisite:

Manufacturing Process, Industrial working environment through industrial training and Industrial visits.

Course Objectives:

1. Understand the principles and evolution of Industrial Engineering, including productivity
1. factors, work study techniques, and their role in process improvement.
2. Apply methods of work measurement, ergonomics, and safety standards to design efficient, safe, and human-centric workplaces aligned with legal and industrial norms.
3. Design optimal facility layouts and evaluate job roles and performance, using tools for facility location, job evaluation, and merit rating to enhance operational efficiency.
4. Implement quality control and assurance practices, and explore recent trends such as TQM, Six Sigma, Kaizen, and ISO standards to improve product and process reliability.

Course Outcomes:

After completing the course, students will be able to

1. Analyze the key principles, tools, and contributors of Industrial Engineering for productivity improvement.
2. Apply work study techniques, ergonomic principles, and safety regulations to design efficient and safe work systems.
3. Evaluate various plant layout strategies and job evaluation methods to recommend improvements in facility planning and human resource allocation.
4. Utilize quality control methods and assess contemporary industrial engineering practices such as TQM, Six Sigma, and ISO standards for performance enhancement.

Unit – I	**Introduction to Industrial Engineering**	**3 Hours**
Definitions and meaning of I.E., objectives of I.E. Productivity- Factors affecting productivity and ways to improve productivity. Work Study–Definitions, objectives, Importance of work study procedure, Relation of work study with – work Simplification, Human Relation		
Unit – II	**Method Study**	**5 Hours**
Definition, objective, Scope of method study, Basic procedure symbols and recording of facts, Charting conventions, Charts–Operation process chart, Flow process chart, Multiple activity chart, Two handed process chart, Diagrams–Flow and string diagram, travel chart Templets and models, Micro motion study. Therbligs SIMO chart, Critical examination and selection, Implementation method		
Unit – III	**Ergonomics and Industrial Safety**	**4 Hours**
Definition, Man Machine system, Types of display, types of control, manual material handling, Anthropometry, Design of work place and working conditions, ILO Norms. Definition of accident, Cause of accident, Prevention of accident, safety measures factor acts, minimum wages act, Employers state Insurance act.		
Unit – IV	**Work Measurements**	**5 Hours**
Definition, objective and techniques of work measurement, time study, stop watch method, performance rating, allowance, relaxation interference contingency, policy, calculation of standard time, work sampling its need and procedure, Predetermined Motion Time Study (PMTS).		
Unit – V	**Facility Locations and Plant Layout**	**5 Hours**
Factors affecting site selection: - Intangible factors for facility location, tangible factor for facility location, advantages and disadvantages of facility location in urban and rural areas. b) Plant Layout: - Characterization of an efficient layout objectives of plant layout, principles of plant layout, procedure in planning layout, types of plant, layout product/line layout, process/functional layout, fixed position/static layout, cellular/Group Technology layout, selection of material handling equipment.		
Unit – VI	**Job Evolutions and Merit Rating**	**2 Hours**
Job evolution: objectives, advantages and procedure, job analysis, job description, job specification, methods of evolution. Merit rating: Objectives and Method of Merit rating.		
Unit – VII	**Quality Assurance**	**3 Hours**
Definition of quality, Quality Control (QC), Quality Assurance (QA), Statistical Quality Control (SQC) and reliability. Importance of quality. Difference between reliability and quality control. Factors affecting and improving reliability. QA tools. Concept of total quality cycle, quality of design, quality of performance, quality of conformity and total quality. Difference between inspection and quality control.		
Unit – VIII	**Recent trends in industrial engineering**	**3 Hours**
International Organization for standardization and its role, ISO standard series and quality managements system, Total Quality Control (TQC) and Total Quality Management (TQM)-philosophical concepts. Concept of six sigma, Kaizen and it's applications.		

ICA consists of minimum 8 practical/assignments from the below list 1. To Study & Prepare Operation Process Chart (OPC) for given assembly. 2. To Study & Prepare Flow Process Chart and Flow Diagram for given assembly for OPC. 3. To study & Prepare Man-Machine Chart for the given situation 4. To study & calculate co-efficient of correlation for time study person using performance rating technique. 5. To study & calculate standard time for given job. 6. Case Study on plant layout 7. Study of existing layout of a workstation with respect to controls and displays and suggesting improved design from ergonomic viewpoint 8. To conduct process capability study for a machine in the workshop/any SSI. 9. Assignment on quality assurance 10. Assignment on recent trends.
Text Books
2. Martand Telsang, Industrial engineering and Production management, S. Chand Publications 3. A. K. Gupta, Engineering Management, S. Chand Publications 4. O. P. Khanna, Industrial Engineering and Management, Dhanpat Rai Publications 5. O. P. Khanna. Work Study, Dhanpat Rai and Sons Publications
Reference Books
International Labour Office, Introduction to work study, Universal Publication
e-Resources
1. https://nptel.ac.in/courses/112107292 2. https://onlinecourses.nptel.ac.in/noc25_me181/preview

Programme Elective Course V
23MAU7E33T: Industrial Robotics

Teaching Scheme		Examination Scheme	
Lectures	2 Hours/week	**ESE**	60 Marks
Practical	2 Hours/week	**ISE**	40 Marks
Credits	3	**ICA**	25 Marks

Introduction:

This course aims to equip students with fundamental knowledge and practical skills in industrial robotics and automation. It covers robot anatomy, configurations, sensors, actuators, end effectors, and applications in industrial environments. The course emphasizes kinematics, dynamics, Jacobian analysis, trajectory planning, and mobile robotics, including AGVs and navigation. Students are also introduced to image processing for robotic vision and robotic work-cell design. Through simulations, case studies, site visits, and hands-on software tools, the course enables students to analyse, design, and apply robotic systems to real-world industrial problems.

Course Prerequisite:

Students enrolling in this course are expected to have prior knowledge of fundamentals of machines and mechanisms, basic differential and integral calculus, matrix algebra, and fundamentals of electrical and electronics engineering. Familiarity with programming using MATLAB, Scilab, or Octave is also required to effectively perform simulations and analytical tasks in the course.

Course Objectives:

1. Understand the basic construction, configuration, and working principles of industrial robots.
2. Analyse the current industrial robot market, its distribution, and emerging future trends.
3. Explain the technologies used in modern robots, including sensors, actuators, grippers, controllers, and machine vision systems.
4. Relate theoretical concepts of industrial robotics to practical industrial applications, thereby bridging the gap between textbook knowledge and real-world industry practices.

Course Outcomes:

After completing the course, students will be able to

1. Apply kinematics and dynamics principles to solve basic robot motion problems.
2. Identify and select appropriate industrial robot specifications based on application requirements.
3. Use software tools to implement and demonstrate fundamental machine vision concepts.
4. Develop and simulate robotic systems and work cells using suitable robot simulation software.

Unit – I	Introduction to Industrial Robots	6 Hours

History and fundamentals of Industrial Robots, Definition as per ISO & IFR, Technology Evolution, components of industrial robots, configuration, typical specifications, current market scenario, “Collaborative Robots”.

Unit – II	**Sensors, Actuators & End Effectors**	**8 Hours**
Sensors: Sensor classification, joint angle sensors, rotary encoders, proximity sensors &switches, range sensors, vision sensors, other sensors for diagnostics, selection of sensors. Actuators: Compare Hydraulic, Pneumatic and Electric drives, Review of DC motors and stepper motors, AC motors, speed control of AC motors, VFD drives, and drive selection criteria. End Effectors: End effectors & grippers, classification, applications, design and selection criteria.		
Unit – III	**Robot Kinematics and Dynamics**	**8 Hours**
Forward kinematics: Coordinate frames, transformations, arm equation. Inverse Kinematics: Tool Configuration, inverse kinematics of 2DOF and 3DOF planar manipulator. Dynamics: Velocity Jacobian, singularities, induced torque and forces, Lagrange's Equation, Dynamic models of two-axis planar robots.		
Unit – IV	**Robot Control and Trajectory**	**6 Hours**
The control problem, state equations, actuator dynamics, set point tracking trajectory planning, joint space schemes, Cartesian space schemes, issues in trajectory planning. Overview of advanced control techniques such as force control, PID control adaptive control etc.		
Unit – V	**Robot Vision/Machine Vision**	**6 Hours**
Machine Vision definition and system components, lighting techniques, Image processing fundamentals: Edge detection, shape analysis, segmentation, object identification, template matching, Cameras (CCD, CMOS, Area Scan, Line Scan), camera specification and selection camera calibration.		
Unit – VI	**Robot Workcells & Programming**	**6 Hours**
Robot cell layout, considerations in workcell design, workcell control, cell safety, human machine interface, robot cell controller. Lead through programming, walk through programming, offline programming.		

ICA consists of minimum 8 Assignment based on curriculum.:

1. Survey assignment on robots industry and manufacturers and applications.
2. Theory assignment on sensors, actuators and grippers
3. One software assignment on kinematic and dynamics.
4. One software assignment robot motor control using Matlab/Scilab/Octave.
5. One assignment on Image Processing using Matlab/Scilab/Octave/LabView.
6. One assignment on Image Analysis using Matlab/Scilab/Octave Labview.
7. Survey assignment on AGVs and mobile robots.
8. One assignment on workcell simulation in **any robot simulation software**. (Robot Studio, RobCAD, Workspace, Delmia IGRIP, V-rep, Gazebo)
9. Trajectory Planning of an Industrial Robot using Matlab/Scilab/Octave (Use image processing toolbox)

Text Books

1. S.K Saha, Introduction to Robotics, McGraw-Hill.
2. Mikell Groover et.al, Industrial Robotics, McGraw Hill.
3. James, Keramas, Robot Technology Fundamentals, Delmar Cengage Learning.
4. Spong & Vidyasagar, Robot Dynamics and Control, Wiley.
5. Gunter Ulrich, Automated Guided Vehicle Systems, Springer.

Reference Books
1. Asitava Ghosal, Robotics: Fundamental Concepts and Analysis, Oxford Press. 2. Siegwart et.al, Autonomous Mobile Robots, Prentice Hall India. 3. Shimon Y. Nof, Handbook of Industrial Robotics, Wiley 4. Schilling, Fundamentals of Robotics, Prentice Hall India. 5. International Federation of Robotics - https://www/ifr.org
e-Resources:
https://www.youtube.com/watch?v=McqDT1DMlh8&list=PLXDsvE7qtfNdt9oYEhJ_LMXDUGu6bH-L6

23MAU7RM4T: Research Methodology and Intellectual Property Rights

Teaching Scheme		**Examination Scheme**	
Lectures	2 Hours/week	**ISE**	50 Marks
Practical	2 Hours/week	**ICA**	25 Marks
Credits	3		

Introduction:

Research Methodology provides a systematic framework for identifying, analyzing, and solving technical problems through evidence-based inquiry. Intellectual Property Rights (IPR) establishes the legal safeguards necessary to protect these engineering innovations, ensuring that creators maintain control and commercial benefits over their inventions. Together, they equip students with the tools to conduct ethical research and navigate the global patent and copyright landscape.

Course Prerequisite:

Basic Engineering Mathematics and Statistics, Basic Science subjects (Physics/Chemistry)
A foundation in technical writing or communication skills

Course Objectives:

1. To introduce the students to the research fundamentals and its systematic process.
2. To develop problem identification and literature review skills.
3. To introduce various research design, sampling, data collection & interpretation methods.
4. To enable the students to identify and apply ethical practices in research including plagiarism avoidance.
5. To know about different forms of Intellectual Property Rights (IPR) and their importance to engineering.
6. To gain practical insights in patent filing, copyright, trademarks, and trade secrets.

Course Outcomes:

After completing the course, students will be able to

1. Differentiate between different types of research and research designs.
2. Conduct literature reviews using online and offline research databases.
3. Formulate research problem statements, hypotheses, and objectives.
4. Use basic statistical techniques for data analysis and interpretation.
5. Choose a proper type of IPR and strategies for IP management for a certain situation

Unit – I	**Introduction to Research**	**6 Hours**

Meaning & objectives of research, Types of research: basic, applied, quantitative, qualitative, Steps in research process, identifying research problems, Formulating hypotheses and objectives

Unit – II	**Literature Review & Research Design**	**4 Hours**

Literature search strategies, IEEE Xplore, Scopus, Google Scholar, Springer, Web of Science; Research gap analysis, Research design: exploratory, descriptive, experimental, Sampling techniques

Unit – III	**Data Collection & Analysis**	**4 Hours**
Primary vs secondary data, Tools of data collection (questionnaires, surveys, interviews), Basics of statistical analysis: Mean, Median, Mode, Standard deviation, correlation; Use of software (Excel, SPSS), Data interpretation and presentation.		
Unit – IV	**Technical Communication & Ethics in Research**	**6 Hours**
Technical writing framework, Research paper structure: Abstract, Introduction, Methodology, Results, Discussion, Conclusion; Referencing and citations (APA/IEEE formats), Ethical considerations & plagiarism, Use of plagiarism checking tools		
Unit – V	**Introduction to Intellectual Property Rights**	**4 Hours**
IPR definitions & relevance to engineering, Patents: types, patentable subject matter; Copyright: scope, ownership, duration; Trademarks, Trade Secrets, Industrial Designs		
Unit – VI	**Patent Rules, Filing, and IP Management**	**4 Hours**
Patent search strategies, Patent drafting basics, Patent filing procedure in India and abroad, Patent Cooperation Treaty (PCT) overview, IPR protection strategy for products and software, Technology transfer, Licensing		

ICA consists of minimum 5 of the following activities should be done in a group of 3-4 students.

1. Identification and Formulation of a Research Problem: Students shall observe a real-life situation (college, community, workplace, social, business, or technical areas etc.), identify one researchable problem, Frame it with Problem statement, Research objectives & at least two hypotheses.

2. Literature Survey and Review using Online Databases: Students shall do Literature Survey and Review using Online Databases like Google Scholar / ResearchGate / journals, Collect 5–7 research papers on the selected topic. They should prepare a literature review table with: Author, Year, Methodology, Findings, Research gap.

3. Sampling Design and Data Collection: Students shall do Sampling Design and Data Collection which will include a designed questionnaire for a selected problem choosing a sampling method (random, convenience, stratified) with data of at least 20–30 respondents.

4. Statistical Analysis: Use statistical software (like SPSS or R) to analyze a dataset, interpreting the results and discussing their implications for the research question.

5. Patent Search and Prior Art Analysis: Students should select a simple invention idea (e.g., eco-friendly bottle, mobile app feature) & conduct prior art search using real databases (Google Patents / Espacenet / IP India). Identify similar patents and analyze novelty.

6. Drafting a Provisional Patent Specification: Using a given or self-chosen invention, students should draft: Title, Field of invention, Background, Summary, Description of invention, Follow, Indian Patent Office provisional format.

7. Identification and Registration Analysis of Geographical Indications: Students should identify a local product (handicraft, agricultural product), Verify GI status from official GI registry. If registered: analyze protection benefits, If not: prepare GI registration feasibility note.

8. Case Study on IPR: Students should select a real IPR case (e.g., turmeric patent, neem patent, software licensing dispute), Analyze: Type of IP involved, Legal issues, Final judgment, Impact on innovation etc.

Text Books

1) Research Methodology: C.R. Kothari , New Age International Publications
2) Research Methodology, A Practical and Scientific Approach - Vinayak Bairagi, Mousami V. Munot, Chapman and Hall
3) Research Methodology: A step-by-step guide for beginners - Kumar, R., SAGE Publications Pvt. Ltd
4) An Introduction to Research: Donald Ary, Pearson Publications
5) Intellectual Property Rights: P. Narayanan, S. Chand Publications
6) Fundamentals of IPR: T. Ramappa, S. Chand Publications

Reference Books

1) Research Methodology: Methods & Techniques, Ranjit Kumar, Sage Publications
2) The Craft of Research, Wayne C. Booth et al., Chicago Press
3) Patent Law and Policy, Amy L. Landers, West Academic
4) Engineering Research Methodology, Nakkeeran, K.K. Publications
5) Intellectual Property Rights: Unleashing Knowledge Economy, Prabuddha Ganguli
6) Intellectual Property Rights under WTO, T. Ramappa
7) Intellectual Property Handbook, WIPO (World Intellectual Property Organization, free PDF).
8) Intellectual Property Rights and Innovation, OECD (Open access reports).

e-Resources

Relevant NPTEL Courses

1) Research Methodology
2) Intellectual Property Rights: A Management Perspective
3) Patent Drafting
4) Technical Communication

Important Websites / E-Resources

1) IEEE Xplore Digital Library
2) Scopus / Web of Science
3) Google Scholar

Patent Databases

1) USPTO (United States Patent Office)
2) WIPO (World Intellectual Property Organization)
3) Indian Patent Advanced Search System (InPASS)

Other MOOC courses:

Coursera / edX / Udemy – Courses on research skills & IPR

23MAU7PR4L: Capstone Project-I

Teaching Scheme		Examination Scheme	
Lectures	----	**ICA**	100 Marks
Practical	8 Hours/week	**OE**	50 Marks
Credits	4		

Introduction:

The Capstone Project-I provides students an opportunity to integrate the knowledge and skills acquired throughout the undergraduate program to solve a real-world engineering problem. Students are expected to identify a relevant industrial, societal, environmental, or research problem and develop an innovative engineering solution using systematic design methodologies and modern engineering tools. Students are encouraged to undertake interdisciplinary projects in collaboration with industries, research organizations, start-ups, or community stakeholders.

Course Prerequisite:

Students should possess fundamental knowledge of the core subjects completed during the previous semesters

Course Objectives:

1. Understand the basic concepts & broad principles of Industrial or social project ideas.
2. Make students aware of various databases/ sources of gathering required data for solving the problem selected by them.
3. Learn the principles of coordination & teamwork

Course Outcomes:

After completing the course, students will be able to

1. Identify and define complex engineering problems through systematic need analysis and literature review.
2. Evaluate alternative engineering solutions considering technical feasibility, cost, safety, sustainability, and societal impact.
3. Design the preliminary architecture, mathematical model, simulation model, or prototype using appropriate engineering tools.
4. Develop a comprehensive project proposal incorporating objectives, methodology, milestones, project management plan, and expected deliverables.

Guidelines for Project content:

Project Term Work: The term work under project submitted by students shall include:

a. Weekly reporting

The project group shall maintain regular progress records, verified periodically by the project guide. The records shall reflect the efforts taken by the group for:

1. Searching suitable project work
2. Brief report, preferably on Journals/ conference papers/ books or literature surveyed to select and bring out the project.
3. Brief report of feasibility studies carried to implement the project objectives.
4. Proposed diagram/ Design calculations, etc.

b. Synopsis:

The group should submit the synopsis (of 5-8 pages) in following form.

1. Title of Project
2. Names of Students
3. Name of Guide
4. Proposed work (Must indicate the scope of the work & weekly plan up to March end)
5. Approximate Expenditure (if any).

The synopsis of project is expected to approve by the guide and endorsed by the Head of the Department.

Note: - The project group consist not more than three students. The group has to submit the report on the progress of project work at the end of first semester.

WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR **(An Autonomous Institute)** **Final Year B.Tech. (Mechanical and Automation Engineering), Semester-VIII**	

23MAU8E21T: Power Plant Engineering

Teaching Scheme		**Examination Scheme**	
Lectures	2 Hours/week	**ESE**	50 Marks
Tutorial	----	**ICA**	----
Credits	2	**ISE**	----

Introduction:

This Course provides a simple understanding of the power plant engineering. The course contains the details of steam and gas thermal power plants, hydro power plants, nuclear power plants, along with solar, wind and geothermal energy power systems in addition to the direct energy conversion. The economics of power generation and the environmental aspect of power generation are also being addressed in this course.

Course Prerequisite:

No prerequisite required

Course Objectives:

1. To understand the global energy scenario and the working principles of various conventional power plants including steam, gas turbine, hydro, and nuclear power plants.
2. To analyze the components and performance of steam power plants such as boilers, turbines, condensers, feed water systems, and fuel handling systems.
3. To study renewable energy systems including solar, wind, geothermal, and ocean energy, along with energy storage techniques.
4. To evaluate the economic and environmental aspects of power generation and compare different power plant technologies

Course Outcomes:

After completing the course, students will be able to

1. Explain the working principles and components of steam, gas turbine, hydroelectric, and nuclear power plants.
2. Analyze the performance of boilers, steam turbines, condensers, and hydraulic turbines through problem solving.
3. Compare conventional and renewable energy power plants based on efficiency, application, and sustainability.
4. Assess the economics and environmental impacts of different power generation methods and recommend suitable power plant systems for given conditions.

Unit – I	**Energy Scenario and Steam Power Plant Systems**	

The energy scenario, steam power plants, fuel handling, ash handling, chimney draught

Unit – II	**Boilers, Combustion Systems and Steam Turbines**	
Fossil fuel steam generators, high pressure boilers, performance of boilers, fuels and combustion, steam turbines		
Unit – III	**Steam Turbine Types and Condenser Systems**	
Impulse turbines, reaction turbines, feed water treatment, steam condensers, problem solving		
Unit – IV	**Gas Turbine, Combined Cycle and Hydroelectric Power Plants**	
Condensate feed water system, circulating water system, gas turbine cycles, combined cycles, hydro-electric, power plants		
Unit – V	**Hydraulic Turbines and Hydro Plant Control Systems**	
Classification of hydro-plants , hydraulic turbines, hydro plant controls, problem solving		
Unit – VI	**Nuclear Power Plants and Solar Thermal Energy Systems**	
Principles of nuclear energy, thermal fission reactors and Power Plants, Fast breeder reactors, solar energy, solar thermal energy		
Unit – VII	**Renewable Energy Systems and Direct Energy Conversion**	
Solar thermal energy, direct energy conversion, wind energy, geothermal energy, energy from oceans		
Unit – VIII	**Energy Storage, Economics and Environmental Aspects of Power Generation**	
Energy storage, economics of power generation, economics of power generation, environmental aspect of power generation, problem solving		

Reference Books

1. Black & Veatch, "Power plant Engineering", CBS Publisher, 2005
2. El-Wakil, M.M., "Power plant Technology", McGraw-Hill Book Co, 2002
3. Nag, P.K., "Power plant engineering", Tata MacGraw Hill, 2008
4. Modern Power Station Practical, CEGB, Pergamon Publisher, 1992
5. Norris & Therkelsen, "Heat Power", McGraw Hill, 1999
6. Rust, J.H., "Nuclear Power Plant Engineering", Haralson Pub. Co., 1999
7. Potter, P.J., "Power Plant Theory & Design", Kreiger Publishing Co., 1994

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https://nptel.ac.in/courses/112107291

23MAU8E42T: Welding Application Technology

Teaching Scheme		**Examination Scheme**	
Lectures	2 Hours/week	**ESE**	50 Marks
Tutorial	----	**ICA**	----
Credits	2	**ISE**	----

Introduction:

Welding Application Technology is a specialized course focused on understanding the principles, methods, and practical aspects of welding in modern manufacturing and engineering applications. The course emphasizes the study of welding residual stresses, distortions, and their mitigation, along with techniques for measurement and analysis of welding-induced stresses. Students will gain knowledge of a wide range of welding processes including Plasma Arc Welding, Flux Cored Arc Welding, Thermit Welding, Resistance Welding, Friction Welding, Friction Stir Welding, Soldering, and Brazing. The course also covers the design and analysis of butt and fillet welded joints under various loading conditions, enabling students to evaluate the strength and performance of welded structures. Through this course, learners will develop both theoretical understanding and practical skills required for efficient and reliable welding applications in industrial settings.

Course Prerequisite:

Manufacturing Processes

Course Objectives:

1. To understand the fundamentals of welding residual stresses, distortions, and techniques for their control and mitigation.
2. To learn methods for measuring and analyzing welding residual stresses and distortions using both destructive and non-destructive techniques.
3. To gain knowledge of various welding processes, including arc welding, resistance welding, friction welding, and allied joining techniques.
4. To develop the ability to design and analyze welded joints for strength and reliability under different loading conditions.

Course Outcomes:

After completing the course, students will be able to
1. Explain the causes, effects, and control methods of residual stresses and distortions in welded structures.
2. Apply measurement and analysis techniques to evaluate welding-induced stresses and distortions accurately.
3. Identify and select appropriate welding processes for different engineering applications.
4. Design and analyze butt and fillet welded joints, ensuring safety and performance under various static and bending loads.

Unit – I	**Basics of welding residual stresses; distortions and its mitigation**	
Residual Stress, Influencing Factors and Control of Residual Stresses		
Unit – II	**Measurement; analysis of welding residual stresses and distortions**	
Residual Stress Measurement, Residual Stress Measurement by NDT		
Unit – III	**Measurement of welding residual stresses and distortions**	
Welding Induced Distortion- Control and Measurement, Measurement and Prediction		
Unit – IV	**Different type of welding methods and its details-I**	
Plasma Arc Welding (PAW), Flux Cored Arc Welding (FCAW), Thermit Welding		
Unit – V	**Different type of welding methods and its details-II**	
Resistance Welding, Types, Friction Welding		
Unit – VI	**Different type of welding methods and its details-III**	
Friction Stir Welding, Soldering, Brazing		
Unit – VII	**Design & analysis of butt and fillet welds joints**	
Strength of parallel and traverse fillet welds, analysis of eccentrically loaded joints,		
Unit – VIII	**Design & analysis of weld joints for different static loading conditions**	
Welded joints subjected to bending moment		

Reference Books

1. V. M. Radhakrishnan, Welding Technology and Design, New age. 2002.
2. Dr. O. P. Khanna, Welding Technology, Reprint: 2002.
3. J. A. Goldak, Computational Welding Mechanics, Springer 2005.
4. O. Grong, Metallurgical Modelling of Welding,2nd Ed. IOM publication , 1997.
5. L-E Lindgren, Computational Welding Mechanics, Woodhead Publishing Limited,2007.
6. J. F. Lancaster (Ed), The Physics of welding, Pergamon, 1986.
7. R.W. Messler, Principles of Welding, John Wiley and Sons,1999.

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https://onlinecourses.nptel.ac.in/noc22_me121/preview

WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR (An Autonomous Institute) Final Year B.Tech. (Mechanical and Automation Engineering), Semester-VIII

23MAU8E52T: Advances in Welding and Joining Technologies

Teaching Scheme		Examination Scheme	
Lectures	2 Hours/week	**ESE**	50 Marks
Tutorial	----	**ICA**	----
Credits	2	**ISE**	----

Introduction:

The progress of several welding and joining processes is ever increasing with the development of new materials and their application in modern technologies. The micro joining and nano joining is even more challenging area with the development of miniature components. This course is primarily designed from fundamental understanding to the most recent advances in welding and joining technologies. The syllabus is oriented to the advancement of the joining technologies which is different from conventional welding and joining processes. The modules cover almost all the direction of joining technologies and it is blended with fundamental development to the recent technologies. Audience will be able to develop fundamental understanding on different perspective and recent development in this field through the lectures and reinforce their knowledge by solving assignments. This course is presented in a lucid and simplified way to make it enjoyable to the beginners.

Course Prerequisite:

No prerequisites required

Course Objectives:

1. To understand the fundamental principles of welding and joining, including fusion, solid-state, brazing, soldering, and adhesive bonding.
2. To study advanced welding techniques such as laser, electron beam, and micro/nano joining for modern industrial applications.
3. To analyze welding mechanics, heat flow, stress, strain, and metallurgical transformations in welded joints.
4. To explore welding and joining of non-metallic materials and emerging technologies such as Cold Metal Transfer and metal printing.

Course Outcomes:

After completing the course, students will be able to

1. Discuss the physics, classifications, and mechanisms of various welding and joining processes.
2. Apply knowledge of laser, electron beam, and solid-state welding techniques to select suitable processes for specific engineering applications.
3. Analyze thermal, mechanical, and metallurgical effects in welded joints using computational and theoretical models.
4. Evaluate advanced joining techniques for metals, plastics, ceramics, and medical components, including micro- and nano-scale applications.

Unit – I	**Fundamentals of welding and joining**	
Classification, physics of welding, fusion welding, Brazing and Soldering, Adhesive Bonding, physics of welding arc, surface active elements and allied welding processes, magnetically impelled arc welding		
Unit – II	**Laser and electron beam welding**	
Principles of laser, types of lasers and their applications, keyhole and conduction mode of laser welding, laser welding of metals, pulse shaping in laser welding, laser assisted hybrid welding, Electron beam welding		
Unit – III	**Solid state welding processes**	
Solid state bonding mechanism, Cold welding, Ultrasonic welding, Friction Stir welding, Diffusion bonding, Explosive welding, Electromagnetic pulse welding		
Unit – IV	**Computational welding mechanics**	
Analysis of heat flow, cooling rates, models for welding heat sources, analytical solution of temperature distribution, heat transfer, fluid flow, influence of surface-active elements, analysis of stress and strain in welding		
Unit – V	**Micro-joining and nano-joining**	
Fundamentals of fusion micro welding, microelectronics wire bonding, bonding using nano particles, Laser micro welding, Electron bean micro welding, Resistance micro welding, Plastics micro welding, micro joining of medical components and devices, advances I laser and other micro welding processes, development of nano joining technologies		
Unit – VI	**Welding metallurgy**	
Chemical reaction in welding, residual stress and distortion, solidification in welding, weld microstructure, phase transformation, solidification cracking, heat treatment of weld joint		
Unit – VII	**Welding and joining of non-metals**	
Plastic welding, Ultra-short pulse laser welding, Laser transmission welding, Glass and ceramic welding, Biocompatible plastic to metal welding		
Unit – VIII	**Metal transfer in welding and metal printing**	
Fundamentals of metal transfer in arc welding, Cold Metal Transfer (CMT), introduction to metal printing		

Reference Books

1. J. F. Lancaster: The Physics of welding, Pergamon, 1986
2. R. W. Messler: Principles of Welding, John Wiley and Sons, 1999.
3. Y. N. Zhou: Microjoining and Nanojoining, Woodhead publishing, 2008.
4. W Steen: Laser Material Processing, Springer-Verlag, 1991.
5. Sindo Kou: Welding Metallurgy, Wiley, 2002
6. J. A. Goldak: Computational welding mechanics, Springer, 2005
7. M-K Besharati-Givi and P. Asadi: Advances in Friction-Stir Welding and Processing, Woodhead Publishing Limited, 2014
8. J. Norrish: Advanced welding Processes, Woodhead publishing, 2006
9. L-E Lindgren: Computational welding mechanics, Woodhead Publishing Limited, 2007

10. Ian Gibson, David W. Rosen, Brent Stucke: Additive Manufacturing Technologies, Springer, 2009.
e-Resources
https://nptel.ac.in/courses/112103244

WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR
(An Autonomous Institute)
Final Year B.Tech. (Mechanical and Automation Engineering), Semester-VIII

23MAU8E62T: Work System Design

Teaching Scheme		**Examination Scheme**	
Lectures	2 Hours/week	**ESE**	50 Marks
Tutorial	----	**ICA**	----
Credits	2	**ISE**	----

Introduction:

Work System Design deals with the systematic examination of the methods of doing work with an aim of finding the means of effective and efficient use of resources and setting up of standards of performance for the work being carried out. The systematic examination of work involves what is done? And how it is done? As well as what is the standard time to do the work? This is required to have an in-depth analysis of all the elements, factors, resources and relationships affecting the efficiency and effectiveness of the work being studied. The course also aims at scientifically establishing the time required for a qualified worker to carry out a work element at a defined rate of working. Ergonomic aspects of work system design are also included in the course contents. The scope of this course is not only limited to the manufacturing applications but it is also relevant for service sector industry.

Course Prerequisite:

No prerequisite required

Course Objectives:

1. To understand the fundamentals of productivity, work system design, and measurement techniques for effective work management.
2. To develop skills in work study, method study, and process charting techniques for analyzing and improving industrial operations.
3. To learn advanced work measurement techniques, including time study, performance rating, work sampling, and micro-motion analysis.
4. To apply ergonomics principles to optimize human-machine systems and improve worker safety, comfort, and productivity.

Course Outcomes:

After completing the course, students will be able to
1. Analyze productivity using measurement models, identify factors affecting performance, and suggest improvement techniques.
2. Apply work study and method study techniques to design efficient workflows and reduce wastage of time and effort.
3. Perform work measurement, time study, and performance rating to determine standard times and optimize operations.
4. Evaluate ergonomics of workstations and human-machine systems, proposing design improvements for safety, comfort, and efficiency.

<table>
<tr><td>Unit – I</td><td>Work System Design</td><td></td></tr>
<tr><td colspan="3">Introduction, Introduction and Concept of Productivity, Measurement of Productivity, Productivity Measures, Productivity Measurement Models</td></tr>
<tr><td>Unit – II</td><td>Productivity Analysis, Measurement and Improvement Techniques</td><td></td></tr>
<tr><td colspan="3">Factors Influencing Productivity, Causes of Low Productivity, Productivity Measurement Models, Productivity Improvement Techniques, Numerical Problems on productivity, Case study on productivity.</td></tr>
<tr><td>Unit – III</td><td>Work Study</td><td></td></tr>
<tr><td colspan="3">Basic Concept, Steps Involved in Work Study, Concept of Work Content, Techniques of Work Study, Human Aspects of Work Study</td></tr>
<tr><td>Unit – IV</td><td>Method Study</td><td></td></tr>
<tr><td colspan="3">Basic Concept, Steps Involved in Method Study, Recording Techniques, Operation Process Charts, Operation Process Charts: Examples.</td></tr>
<tr><td>Unit – V</td><td>Process Charting Techniques and Flow Diagram Analysis</td><td></td></tr>
<tr><td colspan="3">Flow Process Charts, Flow Process Charts: Examples, Two-Handed-Process Charts, Multiple Activity Charts, Flow Diagrams.</td></tr>
<tr><td>Unit – VI</td><td>Work Study Techniques</td><td></td></tr>
<tr><td colspan="3">String Diagrams, Principles of Motion Economy, Micro-Motion Study, Therbligs, SIMO Charts</td></tr>
<tr><td>Unit – VII</td><td>Advanced Work Study and Method Improvement Techniques</td><td></td></tr>
<tr><td colspan="3">Memo-Motion Study, Cycle graph and Chrono-Cycle Graph, Critical Examination Techniques, Development and Selection of New Method, Installation and Maintenance of Improved Methods.</td></tr>
<tr><td>Unit – VIII</td><td>Work Measurement</td><td></td></tr>
<tr><td colspan="3">Basic Concept, Techniques of Work Measurement, Steps Involved in Time Study, Steps and Equipment of Time Study, Performance Rating.</td></tr>
<tr><td>Unit – IX</td><td>Performance Rating</td><td></td></tr>
<tr><td colspan="3">Examples, Allowances, Computation of Standard Time-I, Computation of Standard Time-II, Case Study</td></tr>
<tr><td>Unit – X</td><td>Work Sampling</td><td></td></tr>
<tr><td colspan="3">Basics, Procedure of Work Sampling Study, Numerical Problems on work sampling, Introduction to Synthetic Data and PMTS, Introduction to MTM and MOST</td></tr>
<tr><td>Unit – XI</td><td>Ergonomics</td><td></td></tr>
<tr><td colspan="3">Basic Concept, Industrial Ergonomics, Ergonomics: Anthropometry, Man-Machine System-1 , Man-Machine System-2, Case Study: Office Chair, Case Study: Tower Crane Cabin, Case Study: Car Seat, Case Study: Computer System, Case Study: Assembly Line</td></tr>
</table>

Reference Books
1. Introduction to Work Study: International Labor Office (ILO), Geneva. 2. Ralph M. Barnes, Motion and Time Study Design and Measurement of Work, Wiley, The University of California. 3. M. Telsang, Industrial Engineering and Production Management, S. Chand and Company Ltd.
e-Resources
https://nptel.ac.in/courses/112107249

	WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR **(An Autonomous Institute)** **Final Year B.Tech. (Mechanical and Automation Engineering), Semester-VIII**

23MAU8PR4L: Capstone Project-II

Teaching Scheme		**Examination Scheme**	
Lectures	----	**ICA**	100 Marks
Practical	4 Hours/week	**OE**	50 Marks
Credits	2		

Introduction:

Capstone Project-II focuses on implementation, testing, validation, optimization, and documentation of the engineering solution proposed during Project-I. Students are expected to fabricate, develop, simulate, integrate, or experimentally validate the proposed system and demonstrate its performance against predefined objectives. The course promotes innovation, research publication, intellectual property generation, entrepreneurship, and lifelong learning while strengthening professional ethics, communication, and project management skills.

Course Prerequisite:

Basic knowledge of all the core subjects completed during the previous semesters

Course Objectives:

1. Learn how to prepare a good technical report of the project undertaken
2. Understand how presentation of a technical work should be done before the experts
3. Make aware students about publication of research papers in conferences /journals, patent filing & incubation

Course Outcomes:

After completing the course, students will be able to

1. Implement and integrate engineering components into a functional prototype or validated engineering solution.
2. Evaluate system performance using appropriate testing procedures, statistical analysis, and engineering standards.
3. Demonstrate professionalism, teamwork, ethics, leadership, project management, and lifelong learning during project execution.
4. Prepare a comprehensive technical report and effectively communicate project outcomes through presentations, publications, or demonstrations.

The group has to complete the project work and submit the hard bound copy of the report of the same at the end of this semester. Student are encouraged to publish a research paper concerned with their capstone project work in National/International conference